

PF THEORY

ASSIGNMENT 1

CONDITIONAL STATEMENTS

Name: Zaeena Aftab
Section: BS(AI)-1A
Roll No: 26K-0011

PART A

Q₁

⇒ Pseudocode

Output "Enter A number"

Input Num

WHILE Num $\geq$ 0 DO

 Output Num

 Num $\leftarrow$ Num - 1

ENDWHILE

Q₂

⇒ Pseudocode

Output "Enter a year"

Input year

IF year MOD 4 $\neq$ 0 AND year MOD 400 $\neq$ 0 THEN

 Output "It is a leap year"

ELSE

 Output "Not a leap year"

ENDIF

Q3

⇒ Pseudocode

Output "Enter A number"

Input Num

Num2 $\leftarrow$ 0

WHILE Num > 0 DO

Num $\leftarrow$ Num DIV 10

Num2 $\leftarrow$ Num2 + 1

END WHILE

output "your number has ", digits, " digits"

Q4

⇒ Pseudocode

Num $\leftarrow$ 0 , Num2 $\leftarrow$ 0

WHILE Num $\leq$ 10 DO

Num $\leftarrow$ Num + 1

IF Num MOD 3 $\neq$ 0

WHILE Num2 $\leq$ 10 DO

Answer $\leftarrow$ Num1 $\times$ Num2

output Answer

END WHILE

ENDIF

END WHILE

Q₆

⇒ Pseudocode

Sum $\leftarrow$ 0

output "Enter a number "

input Num

WHILE Num $\geq$ 10

Sum $\leftarrow$ 0

temp $\leftarrow$ Num

REPEAT

Sum $\leftarrow$ Sum + (temp MOD 10)

temp $\leftarrow$ temp DIV 10

UNTIL temp = 0

Output Sum

Num $\leftarrow$ Sum

END WHILE

Q1

⇒ Pseudocode

T-pay $\leftarrow 0$

B-rate $\leftarrow 0$

Hotel_Rn $\leftarrow 0$

Output "Total guests "

Input T-guest

n $\leftarrow 0$

WHILE n < T-guest DO

n $\leftarrow n + 1$

output "Enter your room type "

input type

output "Peak season or off-peak? "

input season

output "Total days of the stay? "

input T-days

IF season = "Peak Season" THEN

IF type = "Standard" THEN

B-rate $\leftarrow 5000$

ELSE IF type = "Deluxe" THEN

B-rate $\leftarrow 8000$

ELSE

$B_rate \leftarrow 12000$

ENDIF

ELSE

IF type = "standard" THEN

$B_rate \leftarrow 3000$

ELSE IF type = "Deluxe" THEN

$B_rate \leftarrow 4000$

ELSE

$B_rate \leftarrow 8000$

ENDIF

ENDIF

IF days > 7 THEN

$T_pay \leftarrow B_rate \times T_days \times 0.85$

ELSE

$T_pay \leftarrow B_rate \times days$

ENDIF

$Hotel_Rev \leftarrow Hotel_Rev + T_pay$

output "the total for the guest is ", T_pay

ENDWHILE

output "the total revenue is, ", $Hotel_Rev$

Q2

=> Pseudocode

Elevator $\leftarrow 0$

output "Enter the floor request"

input floor

output "Enter its floor request"

Input Request

IF request > elevator THEN

output "Elevator is moving up"

ELSEIF request < elevator THEN

output "Elevator is moving down"

ELSE

output "Elevators door open"

END IF

elevator = request

Q3

⇒ Pseudocode

deficiency ← FALSE

Sub ← 0

WHILE Sub ≤ 5 DO

 output "Enter your Marks"

 input marks

 IF marks < 33 THEN

 deficiency ← TRUE

 ENDIF

 total ← total + marks

END WHILE

Avg = total / 5

IF deficiency = TRUE THEN

 output "Failed - subject deficiency"

ELSE

 IF avg ≥ 80 THEN

 output "Distinction"

 ELSE IF avg ≥ 60 THEN

 output "You have passed"

 ELSE

print "failed"

END IF

END IF

Q4

=> Pseudocode

Output "Enter quantity of products purchased "

Input quantity

Output "Enter price per item"

Input unit-price

output "Enter discount percentage"

Input discount

Output "Enter tax percentage"

Input tax

$$\text{Total} = (\text{quantity} * \text{unit-price}) \times (100 - \text{discount}/100)$$

$$\text{Bill} = \text{total} + (\text{total} \times \text{tax}/100)$$

output "The Bill is ", Bill

Q5

⇒ Pseudocode

output "No of Vehicles?"

input num

For Index ← 1 TO Num

eligible ← FALSE

output "Enter vehicle type"

input type

WHILE type <> C OR type <> B OR type <> V

output "Invalid type - enter again"

input type

END WHILE

output "Enter F for faculty, S for students, G for guests"

input category

WHILE category <> F OR category <> S OR category <> G

output "Invalid user - enter again"

input category

END WHILE

output "Enter valid parking permit Y/N "

input permit

WHILE permit <> "Y" OR permit <> "N"

output "Invalid permit - enter again "

input permit

ENDWHILE

IF permit = "N" THEN

output "Emergency vehicle Y/N "

input emergency

WHILE emergency <> "Y" OR emergency <> "N" DO

output "Invalid - re-enter "

input emergency

ENDWHILE

IF emergency = "Y" THEN

eligible ← TRUE

ELSE

output "rejected "

rejected ← rejected + 1

ENDIF

ELSE

eligible ← TRUE

ENDIF

IF eligible = TRUE THEN

IF user = "F" THEN

IF ZoneA < 20 Then

ZoneA $\leftarrow$ ZoneA + 1

accept $\leftarrow$ accept + 1

output "Assigned to ZoneA"

output "Remaining capacity is ", 20 - ZoneA

IF type = "C" THEN

car $\leftarrow$ car + 1

ELSE IF type = "B" then

bike $\leftarrow$ bike + 1

ELSE

van $\leftarrow$ van + 1

ENDIF

ELSE

output "Zone A is full"

rejected = rejected + 1

ENDIF

ELSE IF user = "S" THEN

IF type = "V" then

If zoneC $\leq$ 13 then

zoneC = zoneC + 2

accepted = accepted + 1

van = van + 1

output "Assigned to zoneC"

output "Remaining capacity is ", 15 - zoneC

Else

output "zoneC is full"

rejected = rejected + 1

ENDIF

ENDIF

ELSE

If zoneC $<$ 15 then

zoneC = zoneC + 1

accepted = accepted + 1

output "Assigned to zoneC"

output "Remaining capacity is ", 15 - zoneC

If type = "C" then

car = car + 1

ELSEIF type = "B" then

bike = bike + 1

ELSE

$V_{om} = V_{an} + 1$

ENDIF

ELSE

output "Zone C is full"

$rejected = rejected + 1$

ENDIF

ENDIF

NEXT

output "Total Vehicles ", num

output "Total accepted ", accepted

output "Total rejected ", rejected

output "Total Cars ", Cars

output "Total Vans", Vans

output "Total Bikes", Bike

output "Zone A occupancy. ", ZoneA

output "Remaining. ", 20 - ZoneA

output "Zone B occupancy ", ZoneB

output "Remaining ", 40 - ZoneB

output "Zone C occupancy ", ZoneC

output "Remaining " , $15 - \text{ZoneC}$

IF $\text{ZoneA} \geq \text{ZoneB}$ AND $\text{ZoneA} \geq \text{ZoneC}$ Then

output "Zone A has highest occupancy"

Else If $\text{ZoneB} \geq \text{ZoneA}$ AND $\text{ZoneB} \geq \text{ZoneC}$ Then

output "Zone B has highest occupancy"

Else

output "Zone C has highest occupancy"

END IF

IF $\text{ZoneA} = 20$ AND $\text{ZoneB} = 40$ And $\text{ZoneC} = 15$ then

output "parking lot is full"

ELSE

output "Parking lot is not full"

END IF

Q6

Pseudocode

BEGIN

OUTPUT "Enter your VehicleType"

INPUT V_Type

OUTPUT "What's the Current Charging of your Vehicle?"

INPUT Initial_Battery

OUTPUT "Total Duration for Parking?"

INPUT Duration

OUTPUT "Enter the current time"

INPUT C-Time

OUTPUT "Are you a member?"

INPUT Member // Yes or No Boolean Variable

OUTPUT "Are you a disabled Person?"

INPUT Disabled // Yes or No answer in the variable

INPUT Station // Should already be in system not a question from the user

IF V-Type = H

Output "Charging Unavailable - Parking only"

ELSE

Output "No charging Slot Available"

END IF

IF $V_Type = E$

IF Initial-Battery > 40

Output "Vehicle does not qualify for EV charging"

END IF

ELSE

Required_charging $\leftarrow$ Final-Battery - Initial-Battery

IF Required_charging $\leq$ Initial-Battery

Output "No charging Required"

ELSE

IF Initial-Battery ≤ 15 AND

Required_charging ≥ 80

Priority $\leftarrow 1$

Output "Emergency Charging Priority"

ELSE IF (Disabled person = Yes OR Membership = Yes)

AND (Current-Battery ≤ 30)

Priority $\leftarrow 2$

Output "Priority charging"

ELSE

Priority $\leftarrow 3$

Output "Normal Charging"

END IF

IF (C-Time < 17 00 OR C-Time > 22 00)

IF member = Yes AND Priority = 1

Charge-Unit ← 28 // Due to 20 / off

ELSE

Charge-Unit ← 35

ENDIF

ELSE

IF member = Yes AND Priority = 1

Charge-Unit ← 45 // Due to 10 / off

ELSE

Charge-Unit ← 50

ENDIF

ENDIF

ENDIF

ENDIF

IF Duration <= 2

Parking-Charges ← 200

ELSE IF Duration <= 5

Parking-Charges ← 400

ELSE

Parking-charges ← 700

ENDIF

IF Disabled $\leftarrow$ Yes

Parking_Charges $\leftarrow$ 0

ELSE IF Member $\leftarrow$ Yes

Parking_charges $\leftarrow$ Parking_charges - $\left(\text{Parking_charges} \times \frac{20}{100} \right)$

END IF

IF Duration > 8

output "Long-Stay Warning : Please relocate your vehicle
after charging"

ELSE

output "standard parking Duration"

END IF

OUTPUT V_Type

OUTPUT Initial_Battery

OUTPUT Priority

IF (L-Time < 17 00 OR L-Time > 22 00)

OUTPUT "Non-Peak hour charges applied"

ELSE

output "Peak hours charges applied"

END IF

Charging_Cost $\leftarrow$ Charge_Unit $\times$ Total_Units

Output charging_Cost

OUTPUT Parking_Cost

Total_Cost $\leftarrow$ Charging_Cost + Parking_Cost

OUTPUT Total_Cost

IF (C_Time < 17 00 OR C_Time > 22 00)

IF member = yes

OUTPUT "20 / off on Both Parking and charging cost"

ENDIF

ELSE

IF member = yes

OUTPUT "10 / off on Charging cost and 20 / off on parking cost"

ENDIF

ENDIF

IF Disabled = yes

OUTPUT "Parking cost is free"

ENDIF

END CODE |||